Google Cloud H2S

AI Builder Cup

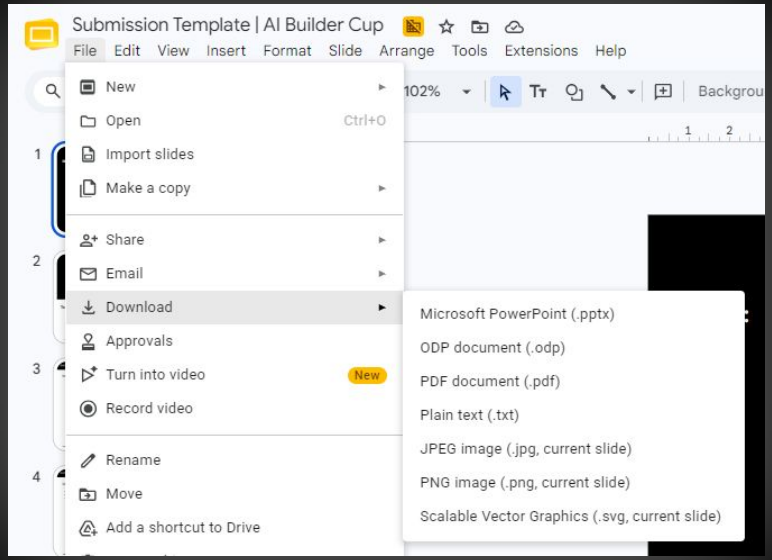
Build in JAPAC.
Build for the world.



Team Details

a. Team name: Market Sense
b. Team leader name: Adwait Kalsekar
c. Problem Statement: El Niño readiness for farmers

Team member: Gursharan Kaur Saini



Note: Please download this template to make your edits and submit the final version.

To download, go to File > Download > Microsoft PowerPoint (.PPTX). Kindly refrain from sending edit requests.

Brief about the idea

SUSTAINABILITY & SOCIAL IMPACT

Kavach — El Niño Readiness Copilot

A global climate warning is not yet a farm decision.

Farmers need to understand local history, limited water and what to verify before sowing.

Our working POC connects sourced environmental evidence, transparent water calculations and simple visual / spoken explanations.

Farmer first: 10 Maharashtra districts · gram, rabi jowar and wheat.

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

DISTINCTIVE APPROACH

Regional El Niño history and visible sources, combined with the farmer's selected crop and water constraints.

PRACTICAL VALUE

Compare two areas against one water budget, see the changed constraint and take a concrete check to a local adviser.

OUR USP

Code calculates; Gemini explains. Visual cards and Hindi voice make evidence easier to understand than a long generic answer.

List of features offered by the solution

IMPLEMENTED

Local context: district, crop, stage, soil and irrigation selections.

Historical comparison: selected El Niño years, rainfall baseline and dated regional surface conditions.

Water options: transparent one-application comparison, selection and printable discussion sheet.

Accessible explanation: contextual chat, visual evidence cards and English / Hindi / Marathi voice.

Guidance: source-linked crop checks, insurance calendar and six-step walkthrough.

Synthetic profiles; regional evidence is not a field measurement or a crop-loss forecast.

Process flow diagram or Use-case diagram

1 Choose context

District, crop and irrigation access

→

2 Explore evidence

Past El Niño years + dated regional conditions

→

3 Compare water

Compare two areas with usable water

4 Ask & understand

Gemini explains the selected evidence in visual cards or Hindi voice.

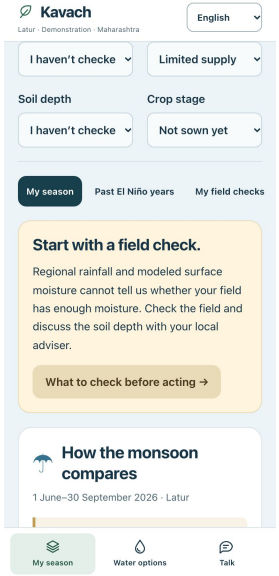
5 Verify before acting

Check field moisture, local agronomic guidance and insurance eligibility.

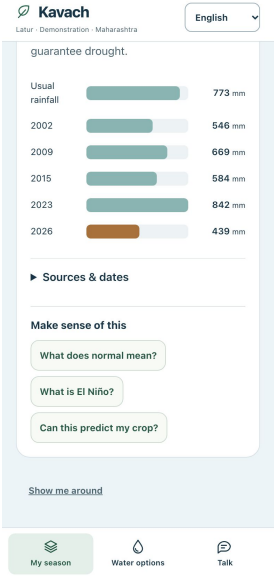
One connected journey: climate context → local history → constraints → next check.

Wireframes/Mock diagrams of the proposed solution (optional)

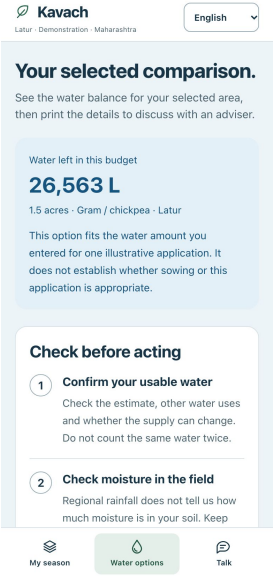
Actual working screens, rather than conceptual mockups.



Choose a situation

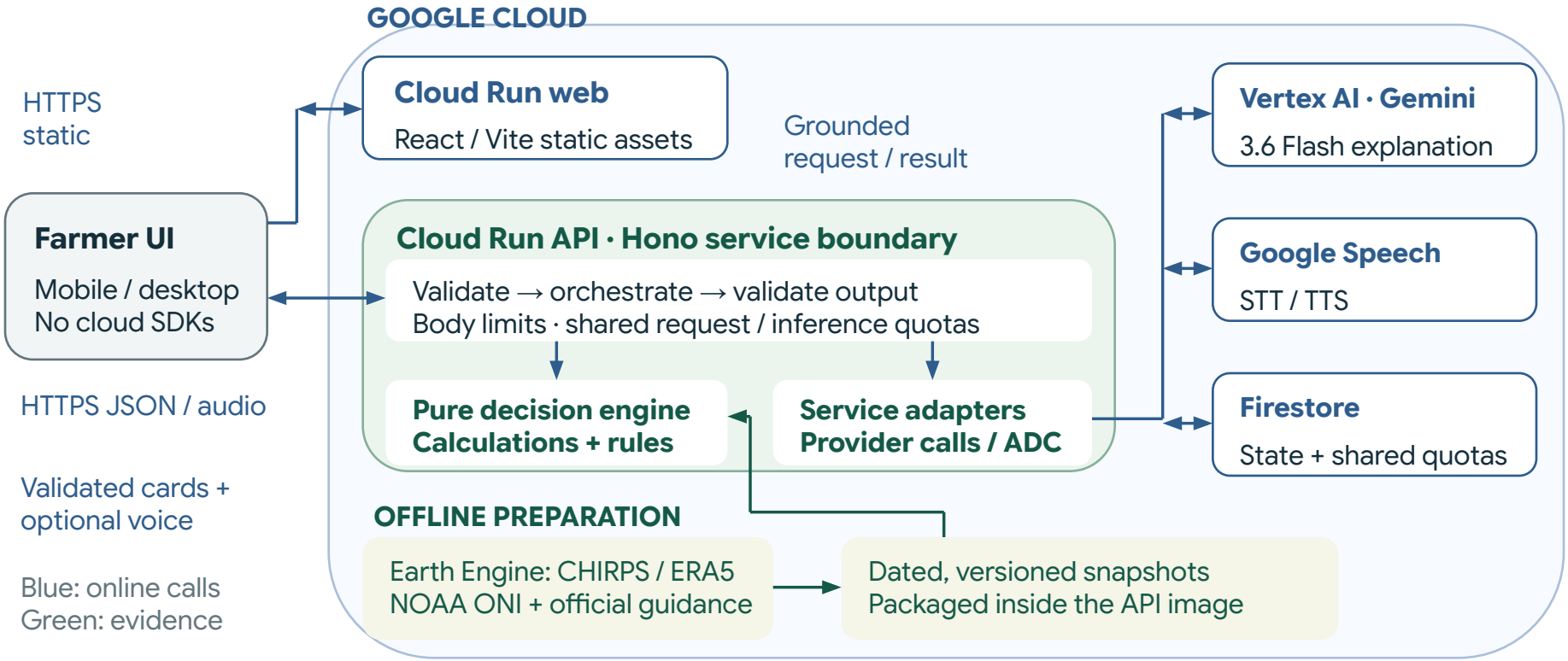


Explore El Niño history



Compare water options

Architecture diagram of the proposed solution



Technologies to be used in the solution

GOOGLE CLOUD

Cloud Run — independent web and API services

Vertex AI — Gemini 3.6 Flash explanation

Google Cloud Speech — STT and TTS

Earth Engine — bounded data acquisition

Firestore — operational state and quotas

APPLICATION & DATA

React, Vite, TanStack Router / Query

Hono, TypeScript and public Zod contracts

Pure deterministic domain calculations

CHIRPS rainfall, ERA5-Land, NOAA ONI

Source-linked official guidance

Implemented stack. BigQuery, Maps and WeatherNext are not part of this POC.

Estimated implementation cost (optional)

Usage-based POC; no measured per-farmer cost yet.

Compute: Cloud Run scales to zero; API capped at one instance.

Inference: shared request quotas, bounded history and response size; no automatic paid retries.

Data: packaged snapshots avoid Earth Engine compute on each user interaction.

Speech: invoked for voice input; text conversations remain silent.

Before a pilot: measure cost per completed journey under realistic traffic.

Trial credits are temporary. Budget alerts are notifications, not a guaranteed billing cap.

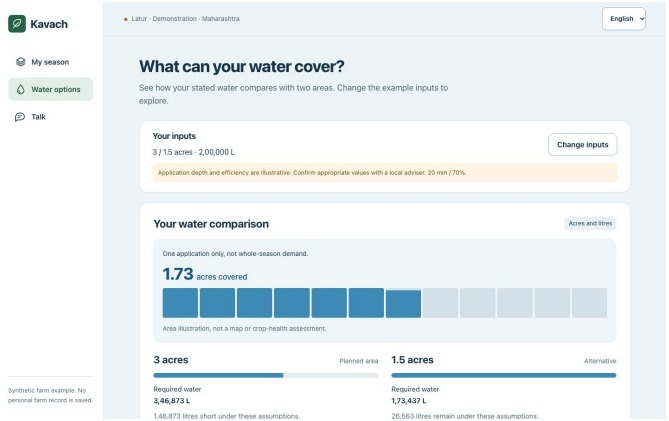
Snapshots of the prototype



Visual chat + follow-ups



Hindi interaction



Desktop experience

Prototype Performance report/Benchmarking

VERIFIED FUNCTIONAL COVERAGE

30 district × crop combinations checked through browser → API.

54 walkthrough checks: 6 steps × 3 languages × 3 viewport sizes.

Conversation: history retained across four turns; distinct one-line / simple rewrites and topic switching.

Real Google speech: English, Hindi and Marathi input / spoken response verified.

12 existing end-to-end cases passed, with build, API, contract and boundary checks.

Engineering verification, not farmer trials. No yield, adoption, loss reduction or latency benchmark is claimed.

Additional Details/Future Development (if any)

NEXT: VALIDATE, THEN EXPAND

Agronomist review: current, dated crop actions and local field validation.

Scientific depth: crop-specific risk calibration, crop-masked vegetation and reservoir access linkage.

Access: fuller numeric voice / localization, village lookup and partner-led farmer trials.

Growth: reusable engine + region packs for new crops, languages and geographies.

Proposed model: free farmer access; institutional API / partnerships with FPOs, advisers and public programmes. No customers or revenue claimed.

Provide links to your:

- GitHub Public Repository
- Demo Video Link (3 Minutes)
- Final Product Link

Repository: https://github.com/adwait-kalsekar/kavach

Video demonstration — https://drive.google.com/file/d/16_JQ9Ign1FwZ1iHbyehvFg2r87MV0ffZ

Live product: https://kavach-web-62041733238.asia-south1.run.app

Evidence and scope

Rainfall: CHIRPS via Earth Engine; monsoon examples 2002, 2009, 2015, 2023; baseline 1991–2020.

Surface context: dated ERA5-Land modeled 0–7 cm moisture / 2 m temperature.

Climate context: NOAA Oceanic Niño Index; history is not a local forecast.

Guidance: official source-linked crop discussion checks and insurance calendars; eligibility remains unverified.

Important limits: town-centre regional grids, synthetic profiles, unreviewed crop prompts; no personal crop-loss prediction or guaranteed sowing recommendation.

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Thank you

